

Miacalcic® Nasal Spray

Regulator of calcium homeostasis.

DESCRIPTION AND COMPOSITION**Pharmaceutical form**

Nasal spray solution in bottles fitted with a metering pump delivering at least 14 doses of 200 IU salmon calcitonin per actuation.

Active substance

The active substance is synthetic salmon calcitonin (INN name calcitonin).

One metered dose delivers 200 IU of synthetic salmon calcitonin.

One International Unit (= IU) corresponds to about 0.2 micrograms of synthetic salmon calcitonin.

Active moiety

Salmon calcitonin.

Excipients

Benzalkonium chloride, sodium chloride, hydrochloric acid (for pH adjustment), water (purified, Eur.P.).

DESCRIPTION

Clear, colourless solution.

INDICATIONS

Miacalcic Nasal Spray is indicated for the treatment of:

Prevention of osteoporosis

In acute bone loss due to sudden immobilization such as in patients with osteoporotic fractures (see PHARMACODYNAMICS). Miacalcic should be supplemented with adequate doses of calcium and vitamin D, as needed by the individual patient, to prevent further loss of bone mass. The maximum duration of treatment is 3 months.

Paget's disease (osteitis deformans):

Only in patients who do not respond to alternative treatments or for whom alternative treatments are not suitable: The duration of treatment is normally 3 months (also refer to DOSAGE AND ADMINISTRATION).

Algodystrophy or Sudeck's disease (neurodystrophic disorders):

Neurodystrophic disorders due to various causes and predisposing factors such as post-traumatic painful osteoporosis, reflex dystrophy, shoulder-arm syndrome, causalgia and drug-induced neurotrophic disorders. The duration of treatment is up to 6 weeks.

DOSAGE AND ADMINISTRATION

It is recommended that the patient alternate between the right and left nostrils as sites of administration for the individual metered doses of nasal spray. In comparison to calcitonin ampoules, the bioavailability of the nasal spray is considerably lower at up to 25%, whereas bioavailability of around 70% is reached with the parenteral solution.

Due to the association between long-term calcitonin use and the occurrence of malignancies (see WARNINGS AND PRECAUTIONS), treatment with calcitonin in all indications should be limited to the shortest period of time possible and using the lowest effective dose.

Osteoporosis

200 IU (1 metered dose of Miacalcic 200) daily, if necessary in several divided doses.

The bioequivalence of the 1 × 200 IU and 2 × 100 IU dosages has not been studied, but data from clinical studies demonstrate the efficacy of both.

Paget's disease

In Paget's disease, the recommended dose is 200 IU daily in two divided doses. 200IU twice daily may be required at the start of treatment in a few cases. Dose reduction may also be attempted during the course of treatment in patients using this route of administration.

The duration of treatment depends on the therapeutic indication and the patient's response. In exceptional circumstances (contraindication of bisphosphonates, severe renal impairment or pathological fractures), treatment may be given for up to 6 months. Thereafter, further treatment is only permissible following careful assessment of the benefits and risks (tumour risk).

Algodystrophy (neurodystrophic disorders)

Early diagnosis of neurodystrophic disorders is important, and treatment should start as soon as the diagnosis is confirmed.

The recommended dose is 200 IU daily in 2 divided doses for 2-4 weeks. Thereafter, 200 IU may be administered three times a week for up to 6 weeks, depending on the patient's clinical response.

Special remarks

Treatment with Miacalcic considerably reduces serum alkaline phosphatase and urinary hydroxyproline excretion, often even to normal levels. Pain is fully or partially alleviated.

In rare cases, alkaline phosphatase and hydroxyproline excretion levels rise after an initial fall. If this happens, the physician must decide on the basis of the clinical picture whether treatment should continue.

Disorders of bone metabolism may recur one to several months after treatment with Miacalcic has been discontinued, necessitating a new course of Miacalcic therapy.

Antibodies to calcitonin may develop in some patients during long-term calcitonin therapy; clinical efficacy is usually not affected, however. Signs of loss of efficacy ("escape phenomenon"), sometimes observed in pagetic patients receiving long-term therapy, are probably due to saturation of the receptors and are apparently not related to the development of antibodies. Following interruption of treatment, the therapeutic response to Miacalcic is restored.

There have been no reports of any pathological changes occurring in the nasal mucosa during long-term treatment with the nasal spray.

Use in children

Miacalcic should not be administered to children for more than a few weeks unless the physician sees compelling reasons for a longer period of treatment. Experience relating to long-term treatment in children is insufficient.

Use in elderly patients / special patient populations

Extensive experience with the use of Miacalcic in elderly patients has shown no evidence of reduced tolerability or of the need for dosage adjustment. The same applies to patients with renal or hepatic impairment, although no specific studies have been carried out in this patient population.

CONTRAINDICATIONS

Known hypersensitivity to synthetic salmon calcitonin or to any of the excipients (see WARNINGS AND PRECAUTIONS and ADVERSE EFFECTS).

WARNINGS AND PRECAUTIONS

Allergic reactions

Since salmon calcitonin is a polypeptide, allergic reactions are possible. Allergic-type reactions, including isolated cases of anaphylactic shock, have been reported in patients receiving Miacalcic. In patients with suspected hypersensitivity to calcitonin, skin testing using diluted, sterile solution from Miacalcic ampoules should be considered prior to initiating treatment.

Risk of malignancies

Meta-analyses of randomized, controlled trials conducted in patients with osteoarthritis and osteoporosis have shown that long-term calcitonin use is associated with a small but statistically significant increase in the incidence of malignancies compared to placebo (see ADVERSE EFFECTS). Patients in these trials were treated with oral or intranasal formulations. The meta-analyses demonstrated an increase in the absolute rate of tumour occurrence in patients treated with calcitonin compared to placebo, which varied between 0.7% (oral formulation) and 2.36% (nasal spray). Numerical imbalances between calcitonin and placebo were observed after 6 to 12 months of therapy. A mechanism for this observation could not be identified. The benefits for the individual patient should be carefully evaluated against possible risks (see ADVERSE EFFECTS).

INTERACTIONS

Concomitant use of calcitonin and lithium may lead to a reduction of up to 30% in plasma lithium concentrations. The dose of lithium may need to be adjusted.

PREGNANCY/BREAST-FEEDING

Women of child-bearing potential

There are no data suggesting that there should be any special recommendations for women of child-bearing potential.

Pregnancy

Reproductive toxicity studies in animals have shown that Miacalcic is devoid of embryotoxic and teratogenic potential. However, there have been no controlled studies and there are insufficient data documented for Miacalcic Nasal Spray in pregnant women. For this reason, Miacalcic should be used with caution during pregnancy.

Breast-feeding

Since there are insufficient data document for Miacalcic Nasal Spray in breast-feeding women, breast-feeding is not advised during treatment. It is not known whether Miacalcic is excreted in breast milk.

Fertility

There are no human fertility data for Miacalcic.

EFFECTS ON ABILITY TO DRIVE AND USE MACHINES

No studies exist on the effects of Miacalcic on the ability to drive and use machines. Miacalcic may cause transient fatigue, dizziness and visual disturbances, which may impair the patient's reactions. Patients must therefore be warned that these effects may occur, in which case they should not drive or use machines.

ADVERSE EFFECTS

Frequencies

Very common ($\geq 1/10$); common ($\geq 1/100$, $< 1/10$); uncommon ($\geq 1/1,000$, $< 1/100$); rare ($\geq 1/10,000$, $< 1/1,000$); very rare ($< 1/10,000$), including isolated reports; in post-marketing use: frequency not known.

Immune system disorders

Rare: Hypersensitivity.

Very rare: Anaphylactic and anaphylactoid reactions, anaphylactic shock.

Metabolism and nutrition disorders

Frequency not known: Hypocalcaemia.

Nervous system disorders

Common: Headache, dizziness, dysgeusia.

Frequency not known: Tremor.

Eye disorders

Uncommon: Visual impairment.

Vascular disorders

Common: Flushing.

Uncommon: Hypertension.

Respiratory disorders

Very common: Nasal discomfort, nasal congestion, nasal oedema, sneezing, rhinitis, nasal dryness, allergic rhinitis, nasal irritation, nasal odour, nasal mucosal erythema, mucosal excoriation.

Common: Epistaxis, sinusitis, ulcerative rhinitis, pharyngitis.

Uncommon: Cough.

Gastrointestinal disorders

Common: Nausea, diarrhoea, abdominal pain.

Uncommon: Vomiting.

Skin and subcutaneous tissue disorders

Rare: Generalized rash.

Musculoskeletal disorders

Common: Arthralgia.

Uncommon: Musculoskeletal pain.

General disorders and administration site reactions (see Respiratory disorders)

Common: Fatigue.

Uncommon: Influenza-like symptoms, oedema (facial, peripheral or generalized).

Rare: Pruritus.

Meta-analyses of randomized, controlled trials conducted in patients with osteoarthritis and osteoporosis have shown that long-term calcitonin use is associated with a small but statistically significant increase in the incidence of malignancies compared to placebo. A mechanism for this observation could not be identified (see WARNINGS AND PRECAUTIONS).

OVERDOSE***Symptoms***

Depending on the dose, parenteral administration may give rise to nausea, vomiting, flushing and dizziness. Such effects might therefore also be expected to occur in association with an overdose of Miacalcic Nasal Spray. However, Miacalcic Nasal Spray has been administered at up to 1,600 IU as a single dose and up to 800 IU per day for three days without causing any serious adverse effects. Isolated cases of overdose have been reported.

Treatment

Management of overdose should be symptomatic.

PHARMACODYNAMICS

All calcitonins consist of 32 amino acids in a single chain, with a ring of 7 amino acids at the N-terminus that differs in sequence from species to species. Salmon calcitonin is more potent and longer-acting than calcitonins from mammalian species due to its greater affinity for receptor binding sites.

Salmon calcitonin inhibits the activity of osteoclasts via their specific receptors. It markedly reduces, and may even normalize, bone turnover in conditions with an increased rate of bone resorption, such as osteoporosis. Salmon calcitonin has been shown both in animal models and in humans to have analgesic activity, presumably via a direct effect on the central nervous system.

Clinical efficacy

Miacalcic produces a clinically relevant biological response in humans after only a single dose, as shown by an increase in the urinary excretion of calcium, phosphorus and sodium (by reducing their tubular reuptake) and a decrease in the urinary excretion of hydroxyproline.

Controlled studies with smaller patient collectives with Miacalcic Nasal Spray have shown that there is a significant reduction in markers of bone turnover such as serum C-telopeptides (sCTX), osteocalcin and skeletal isoenzymes of alkaline phosphatase, at least for the first three months.

Miacalcic produces beneficial effects with a bone-stabilizing effect particularly in postmenopausal women with high bone turnover, alongside analgesia, which is especially beneficial in osteoporotic fractures.

A meta-analysis published in mid-2011, which focussed on pain prevention following vertebral compression fractures (data with nasal spray and ampoules), found significant effects for calcitonin compared to placebo in the first 4 weeks in resting state. For mobile groups, there was also still a small (but statistically significant) effect after 6 months.

In a collective of 467 patients (10 studies), the average age was about 67 years and 90% were women; these data thus show that efficacy is demonstrated especially in postmenopausal women, while use in men and younger patients is only documented in a limited manner.

Calcitonin inhibits gastric and exocrine pancreatic secretion.

PHARMACOKINETICS

Intranasal administration

Various authors have provided divergent data on bioavailability, which is likely to be 25% maximum. As is the case with other polypeptide hormones, plasma levels of salmon calcitonin are not predictive of therapeutic response.

PRECLINICAL SAFETY DATA

Daily intranasal administration for 26 weeks of a placebo solution containing 0.01% benzalkonium chloride or of high doses of a calcitonin formulation containing 0.01% benzalkonium chloride was well tolerated by monkeys. No treatment-related changes to the respiratory tract were observed. Daily intranasal administration of salmon calcitonin with 0.01% benzalkonium chloride to dogs for 4 weeks did not reveal any relevant abnormal findings in the nasal cavity or upper respiratory tract. Miacalcic Nasal Spray with 0.01% benzalkonium chloride did not change nasal ciliary beat frequency following 4 weeks of use in guinea pigs or 6 months of treatment in pagetic patients.

Toxicological findings from long-term studies are attributable to the pharmacological action of salmon calcitonin. Salmon calcitonin is devoid of embryotoxic, teratogenic and mutagenic potential. Toxicity and carcinogenicity studies in rats have shown that salmon calcitonin increases the incidence of pituitary tumours. Further preclinical studies, particularly a mouse carcinogenicity study in which the maximum exposure was more than 7,000 times greater than that in humans following a dose of 200 IU, suggest that this elevated incidence of pituitary tumours is species-specific to rats.

In vivo preclinical safety data do not indicate any association of salmon calcitonin treatment with malignancies and do not provide any evidence for tumour progression.

INCOMPATIBILITIES

None

PACK SIZE

The device consists of a clear, uncolored glass bottle (glass type I) and a spray mechanism with an integrated, automatic dose-counting mechanism and a built-in mechanical stop. The pack contains one bottle with 2 mL of nasal spray solution, fitted with a metering pump delivering at least 14 doses of 200 IU.

STORAGE

See also folding box.

Miacalcic Nasal Spray bottles must be stored at 2 to 8°C. Do not freeze.

Once opened (see section INSTRUCTIONS FOR USE AND HANDLING), they must be kept at room temperature (not above 25°C) and used within a maximum of 4 weeks.

Keep the bottle upright at all times to reduce the risk of air bubbles entering the dip tube.

Miacalcic Nasal Spray should not be used after the date marked “EXP” on the pack.

Miacalcic Nasal Spray must be kept out of the reach and sight of children.

INSTRUCTIONS FOR USE AND HANDLING

The instructions for use and handling of the Miacalcic Nasal Spray included in the section INFORMATION FOR PATIENTS must be read carefully before the spray is used for the first time.

The pump must be primed before first use: Pull up the protective cap, holding the bottle in an upright position, press down the upper part until it clicks. Repeat twice. After the first time the dose counter window shows white and red lines, after the second time white, and after the third time green. It is now ready for use.

INFORMATION FOR PATIENTS

When and how to use Miacalcic Nasal Spray

This medicine is for administration into the nostrils only. You should switch between each nostril every time you use Miacalcic Nasal Spray.

Your pharmacist has stored Miacalcic Nasal Spray in a refrigerator. Before you start using it, you should let the spray reach room temperature.

Instructions for use/handling of Miacalcic Nasal Spray

- Please read the instructions carefully so that you know how to use your nasal spray.
- If the spray mechanism should become blocked, this may be resolved by pressing

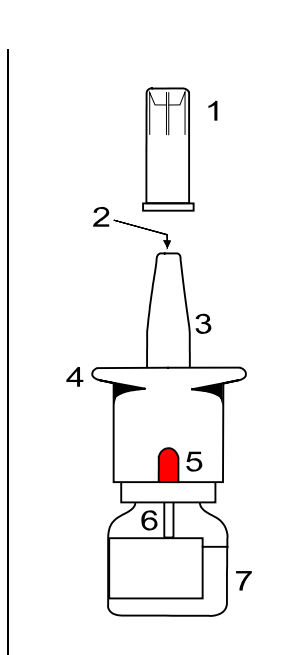
These instructions tell you about:

- The different parts of the nasal spray
- How to prepare a new nasal spray for use
- Using the nasal spray

down firmly on the pump; do not attempt to unblock it by using a sharp pointed object as this may cause damage. If you think your nasal spray is not working properly, take it back to your pharmacist. **Never** try to fix the nasal spray yourself or take it apart, as this may affect your dose.

- Always follow your doctor's instructions regarding dosage carefully.
- Keep this leaflet in a safe place so that you can read it again.

The different parts of your nasal spray

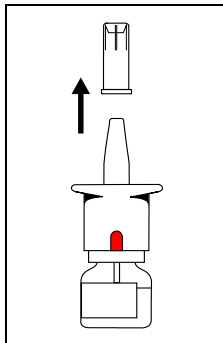


1. **Protective cap:** Keeps the nozzle clean and protects the jet. Always replace the protective cap after you have used the nasal spray.
2. **Jet:** The tiny hole from which the solution sprays out.
3. **Nozzle:** The part you insert into your nostril.
4. **Pump:** The part you press down to operate the spray.
5. **Counter:** The dose counter window on a new nasal spray shows , as shown in this picture. The display will change as you use the pump (see below).
6. **Dip tube:** The tube inside the spray bottle that draws up the solution when you press the pump.
7. **Bottle:** Contains enough solution for at least 14 doses.

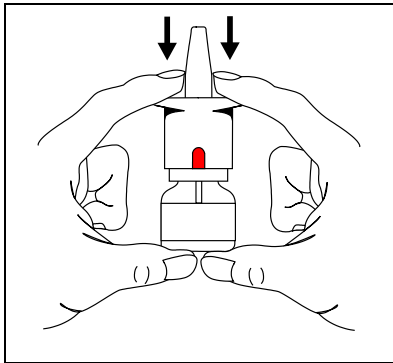
Preparing a new nasal spray bottle for use

NEVER shake the nasal spray bottle as this could cause air bubbles, which may affect your dose.

The dose counter window of a new nasal spray bottle is in the position marked  as shown in the picture



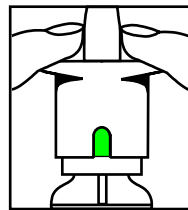
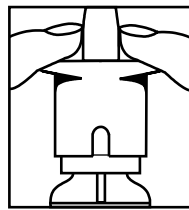
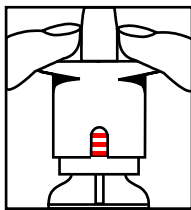
- First, remove the protective cap.



- Hold the nasal spray upright in one or two hands and press down firmly on the pump **3 times**.

This primes the new spray by clearing air out of the dip tube. You will only need to do this once when you start a **new** spray.

Do not worry if a little solution sprays out; this is normal.



- As you press the pump, watch the changes in the dose counter window.
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- When green is showing in the dose counter window, your new nasal spray is ready to use.
- Follow the instructions for 'Using your nasal spray'.

Using your nasal spray

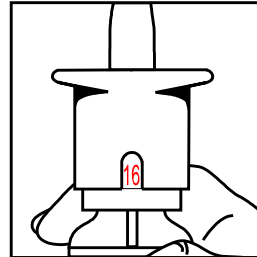


- With the protective cap removed, bend your head forward slightly and insert the nozzle into one of your nostrils. Try to hold the nasal spray upright, as shown in the picture.
- Press the pump firmly **once only**.
- Remove the nasal spray from your nose and breathe in deeply through your nostril to help keep the solution in your nose.
- If your doctor has told you to take two puffs at a time, repeat this procedure in your other nostril.
- After use, clean the nozzle with a dry tissue and replace the protective cap.

Checking the counter:

Every time you use the nasal spray the number in the dose counter window will change. The number displayed tells you how many puffs you have already taken. The nasal spray is guaranteed to deliver 14 metered doses. You may be able to obtain 2 extra doses.

When the dose counter window shows a red 16 as shown in this picture, 16 puffs have been used and the nasal spray is finished. You may notice that a little liquid is left in the nasal spray bottle, but this is normal.



If you are not sure how to use your spray bottle, ask your doctor or pharmacist.

Product Registration Holder:

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Malaysian Package Leaflet

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